

Penalised Maximum Likelihood for Interval-Censored Survival Data

The **survivalMPL** R package

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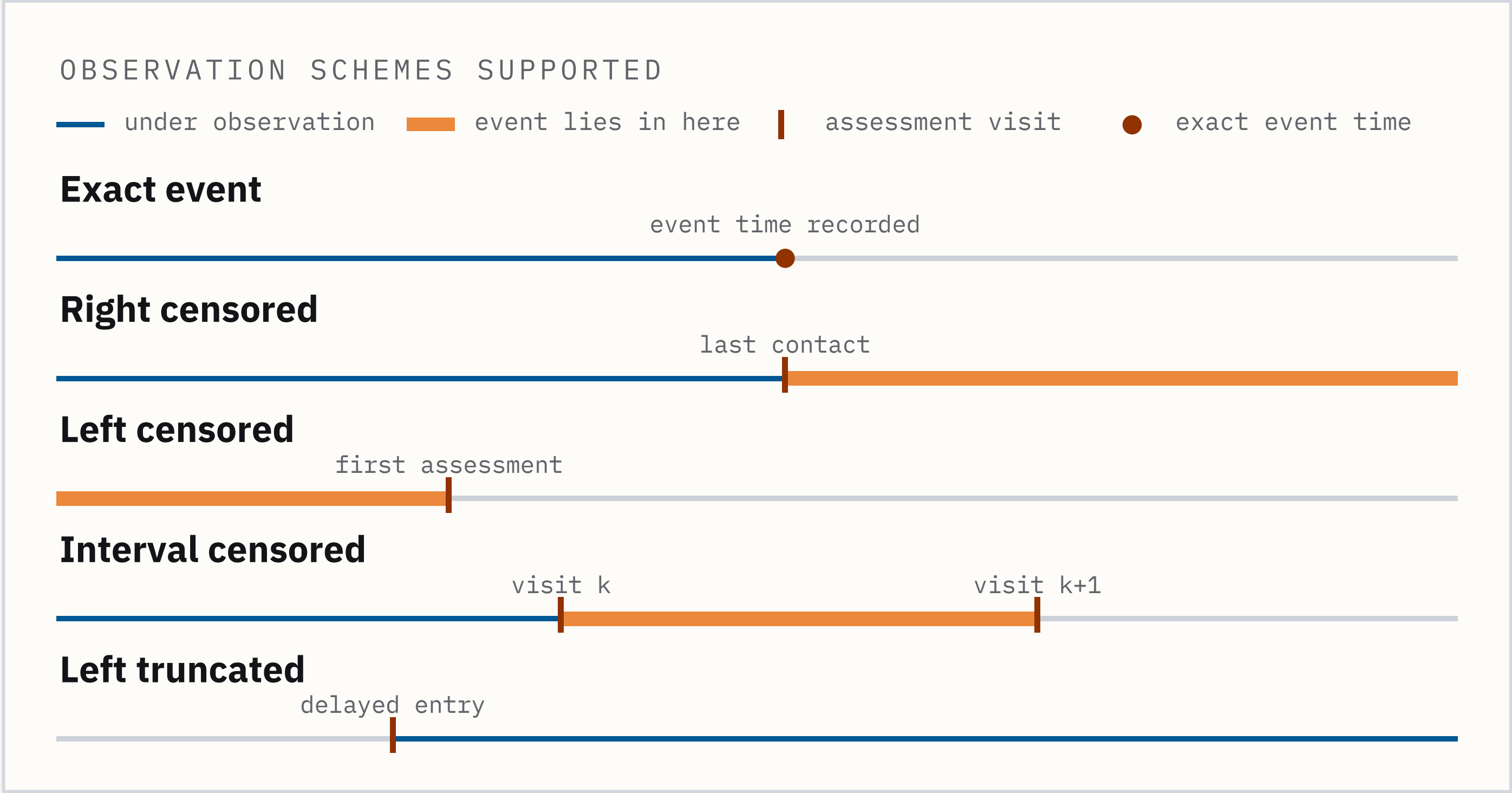
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THE PROBLEM

Trials observe visits, not events. The event time is known only to lie between two assessments.

Most survival software handles this by discarding the interval or collapsing it to a single imputed time. **survivalMPL** fits a Cox model to exact, right-, left- and interval-censored data in one likelihood, with optional delayed entry, and returns a smooth estimate of the baseline hazard rather than a step function.

Where an interval is replaced by a single imputed time, that imputation is then treated as if it had been observed, so standard errors are understated and bias increases with the width of the interval. Maximum penalised likelihood instead considers each interval as recorded and estimates the baseline hazard jointly with the regression coefficients.



4

basis families for the baseline hazard

5

observation schemes, one likelihood

6

S3 methods for fitted objects

7

worked tutorials on the docs site

ARCHITECTURE

Four separated stages, one stable user-facing API.

Model specification

formula · Surv(type = "interval2") · entry=

Estimation

coefficients via penalised likelihood with non-negativity constraints

Inference

standard errors, confidence intervals for coefficients, confidence bounds for hazard-related functions

Post-processing

print · summary · predict · residuals · plot

A WORKED EXAMPLE

bcos2: breast cosmesis after radiotherapy

1 THE CALL

library(survivalMPL)
fit <- coxph_mpl(Surv(left, right, type = "interval2") ~ treatment,
data = bcos2, basis = "msplines")

2 WHAT IT RETURNS

> summary(fit)\$Beta
Estimate Std. Error z-value Pr(>|z|)
treatmentRadChem 0.89355 0.29414 3.03787 0.00238

HR 2.5% 97.5%
treatmentRadChem 2.44 1.37 4.35

> predict(fit, type = "survival", i = 47, time = c(24, 36))
time survival se low high
24 0.4499 0.0574 0.3351 0.5648
36 0.2273 0.0461 0.1351 0.3194

3 WHAT IT MEANS

Adding chemotherapy raises the instantaneous rate of breast retraction by a factor of 2.44 (95% CI 1.37 to 4.35) compared to the radiotherapy-only group. For a woman given both, the fitted probability of remaining free of deterioration is 45% (34 to 56) at two years and 23% (14 to 32) at three.

DOCUMENTATION & REPRODUCIBILITY

A **pkgdown** site with worked examples.

- Getting-started vignette

Left truncation, profile likelihood

coxph() vs coxph_mpl()
- Right, left, interval censoring

Bases and control parameters

Searchable, cross-linked reference

BASELINE HAZARD BASES

The baseline hazard is defined as a non-negative mixture of basis functions.

uniform

Piecewise-constant function. The package default, and the only basis that supports delayed entry.

msplines

Cubic M-splines, the usual recommendation.

gaussian

Truncated Gaussian kernels.

epanechnikov

Compact parabolic kernels.

uniform

msplines

gaussian

epanechnikov

SMOOTHING & METHODS

λ, the penalisation parameter, is selected automatically by marginal likelihood or held fixed at a value user supplies. Non-negativity of θ, the baseline hazard estimates, is a constraint inside the optimiser, not a correction applied afterwards.

coef()

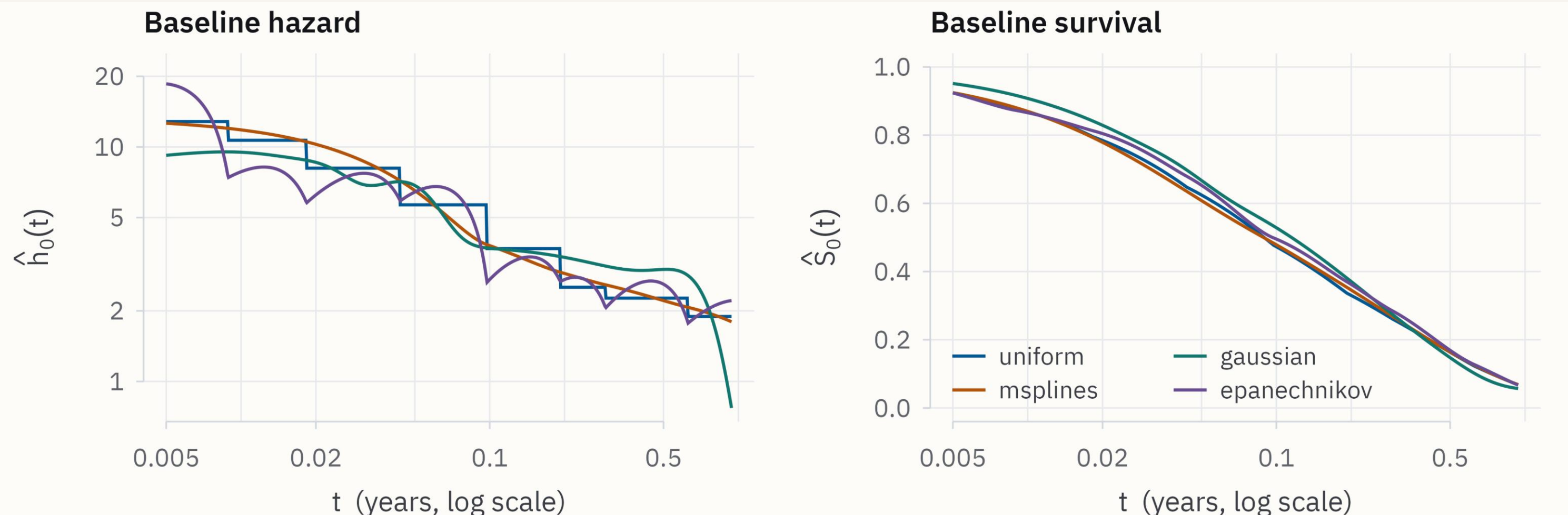
summary()

predict()

residuals()

plot()

print()



IN PRACTICE

By combining our rigorous penalised likelihood framework with an extensible and openly documented implementation, **survivalMPL** brings exact, right-, left- and interval-censored observations, and delayed entry, into a single analysis rather than a collection of workarounds. The worked examples and reference documentation lower the barrier for clinicians, trialists and applied statisticians, and reduce reliance on ad hoc approaches that discard or misrepresent the censoring the trial actually observed.



Full documentation
<https://tinyurl.com/survivalMPL>

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